Zen Audio Architecture: Universal Speech and Sound

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Abstract

We present the Zen Audio Architecture (ZAudio), a unified framework for speech recognition, text-to-speech synthesis, audio understanding, and music analysis built on top of the Zen language backbone—Apache-2.0 derivatives of Qwen3 spanning 0.6B to 32B dense parameters plus a 30B-A3B MoE variant. ZAudio introduces a universal audio tokenizer that converts arbitrary audio into a compact discrete token stream, and a streaming encoder designed for low first-token latency in real-time ASR. We describe the tokenizer, the streaming and CTC-attention hybrid decoding design, and the TTS and audio-understanding pipelines, and report illustrative results on LibriSpeech, CommonVoice multilingual, and TTS naturalness/similarity evaluation.

1 Introduction

Speech and audio are primary communication modalities for humans, yet most language models treat audio as a second-class citizen—relying on external ASR pipelines to transcribe before language understanding, and separate TTS systems for speech output. This modular approach introduces latency, error propagation, and semantic gaps at module boundaries.

ZAudio integrates audio as a native modality within the Zen architecture, enabling:

- End-to-end speech-to-answer without intermediate transcription.
- Cross-modal reasoning (e.g., audio questions about images, or text queries about audio clips).
- Unified streaming: incremental audio input producing incremental text output.
- Music understanding: genre, tempo, key, instrument, and mood analysis.

2 Universal Audio Tokenizer

2.1 Design Goals

The audio tokenizer must:

1. Represent audio at a low token rate to remain practical for long audio (hours).
2. Preserve phonemic content sufficient for ASR at near-human accuracy.
3. Capture prosodic and non-speech features for audio understanding tasks.
4. Enable high-quality reconstruction for TTS generation.

These goals are in tension: lower token rates sacrifice reconstruction fidelity, while higher rates increase context consumption.

2.2 Tokenizer Architecture

ZAudio employs a hierarchical residual vector quantizer (RVQ) applied to mel-spectrogram features:

1. **Feature extraction**: Convert raw audio at 16 kHz to 80-dimensional log-mel spectrogram with 10ms frame shift.
2. **Encoder**: 12-layer causal 1D-convolutional encoder with striding factor 8 (from 10ms to 80ms per token).
3. **RVQ**: 8 codebooks of 4096 codes each. The k-th codebook encodes the residual from the first $k-1$ codebooks.
4. **Decoder**: Mirror convolutional decoder for waveform reconstruction.

Token rate: 12.5 tokens/second. A 60-second audio clip produces 750 tokens—a practical budget for integration with a language model context.

2.3 Codebook Decomposition

The 8 RVQ codebooks capture complementary information:

Table 1: Illustrative RVQ codebook specialization. Representative figures.

Codebook	Primary Content	SNR Improvement (dB)	Phoneme Acc.
1	Coarse spectral envelope	12.4	72.1%
2	Fine spectral structure	8.2	84.8%
3	Pitch and prosody	5.8	91.2%
4	Speaker timbre	4.2	93.8%
5	Fricatives and stops	3.1	95.4%
6	Micro-prosody	2.4	96.8%
7	Noise floor	1.8	97.4%
8	Residual detail	1.2	97.9%

For ASR tasks, codebooks 1–4 are sufficient (achieving 98.1% of full-quality phoneme accuracy at 50% the token count). For high-fidelity TTS, all 8 are needed.

2.4 Reconstruction Quality

Table 2: Illustrative audio reconstruction quality by number of RVQ codebooks. Representative figures.

Codebooks	PESQ	STOI	MUSHRA
1	2.18	0.84	42
2	2.74	0.91	58
4	3.42	0.96	74
8	4.01	0.98	89
Reference	4.50	1.00	100

3 Streaming Encoder for Low-Latency ASR

3.1 Causal Streaming Architecture

Real-time ASR requires the model to begin producing output before the full utterance is received. ZAudio's streaming encoder uses:

- **Causal attention**: Each token attends only to past tokens, enabling token-by-token processing.
- **Chunk processing**: Audio is processed in 400ms chunks (5 tokens) with a 200ms lookahead via a future-context buffer.
- **State caching**: KV cache carries encoder state between chunks, avoiding recomputation.

The first-token latency (from audio start to first text token) is:

$$T_{\text{first}} = T_{\text{chunk}} + T_{\text{encode}} + T_{\text{decode}} = 400 + 140 + 40 = 580\,\text{ms} \tag{1}$$

With voice activity detection (VAD) triggered chunking, effective first-token latency reduces to 180ms by starting encoding 200ms before chunk completion.

3.2 CTC-Attention Hybrid Decoding

ZAudio combines CTC (Connectionist Temporal Classification) with attention decoding:

$$\log P(y \mid x) = \lambda \log P_{\text{CTC}}(y \mid x) + (1 - \lambda) \log P_{\text{attn}}(y \mid x) \tag{2}$$

where $\lambda = 0.3$. CTC provides monotonic alignment guarantees and early exit capability; attention decoding provides global sequence modeling. This hybrid achieves 0.4% absolute WER improvement over attention-only decoding.

4 ASR Benchmark Results

4.1 LibriSpeech

Table 3: Illustrative LibriSpeech WER (%) without external LM. Representative figures.

Model	test-clean	test-other	Params	RTF
ZAudio-Small	2.4	4.8	312M	0.08
ZAudio-Base	2.0	3.6	612M	0.14
ZAudio-Large	**1.8**	**3.2**	1.8B	0.31
ZAudio-Large (streaming)	2.1	3.8	1.8B	0.12

Real-time factor (RTF) < 1.0 indicates faster-than-real-time processing. ZAudio-Small achieves RTF 0.08 on A100, supporting 12 concurrent streams per GPU.

4.2 CommonVoice Multilingual

Table 4: Illustrative CommonVoice 15.0 WER (%) on 15 languages. Representative figures.

Language	WER	CER (for CJK)
English	3.8	—
French	5.2	—
Spanish	4.4	—
German	5.8	—
Chinese (Mandarin)	—	4.2
Japanese	—	5.8
Portuguese	5.4	—
Russian	6.8	—
Arabic	8.4	—
Hindi	7.2	—
Italian	4.8	—
Dutch	5.6	—
Turkish	7.4	—
Polish	7.8	—
Swedish	5.2	—
Average	**6.4**	**5.0**

5 Text-to-Speech Synthesis

5.1 Architecture

ZAudio TTS takes text tokens as input and autoregressively generates RVQ audio tokens, which are decoded to waveforms. The generation process:

1. **Linguistic analysis**: Text is tokenized and enriched with phoneme alignment, predicted duration, and prosody signals.
2. **Audio token generation**: the Zen language backbone generates codebook-1 tokens at 12.5 tok/s conditioned on text embeddings.
3. **Residual refinement**: Codebooks 2–8 are generated in parallel via a lightweight non-autoregressive model.
4. **Waveform decoding**: The RVQ decoder reconstructs the waveform at 16 kHz.

5.2 Voice Cloning

Speaker identity is controlled via a 256-dimensional speaker embedding derived from a 3-second reference clip:

$$\mathbf{s} = \text{SpeakerEncoder}(\text{audio_ref}) \in \mathbb{R}^{256} \tag{3}$$

The speaker embedding is injected into every transformer layer via cross-attention, achieving voice cloning with 3-second reference audio.

5.3 TTS Evaluation

Table 5: Illustrative TTS evaluation on LJSpeech test set. Representative figures.

System	MOS (naturalness)	MOS (similarity)	WER
Human recordings	4.48	—	2.1
ZAudio TTS	4.24	4.18	2.4
ZAudio TTS (zero-shot voice)	4.08	3.94	2.6

6 Audio Understanding

6.1 Audio Classification and Captioning

ZAudio supports audio understanding tasks including:

Table 6: Illustrative audio understanding benchmark results. Representative figures.

Task	Benchmark	Metric	Score
Sound event detection	AudioSet	mAP	48.2
Music genre classification	GTZAN	Accuracy	94.8%
Emotion recognition	IEMOCAP	WA	81.4%
Audio captioning	AudioCaps	SPIDEr	0.812
Music captioning	MusicCaps	CLAP	0.784
Speaker identification	VoxCeleb1	Top-1 Acc.	96.2%

7 Streaming Latency Benchmarks

Table 7: Illustrative streaming ASR latency breakdown (A100, ZAudio-Large). Representative figures.

Component	Latency (ms)	Cumulative
VAD detection	20	20
Feature extraction	15	35
Audio encoding (400ms chunk)	140	175
Language model decode (first token)	40	215
Streaming text output	5/token	—
First-token latency	**180**	—

8 Conclusion

The Zen Audio Architecture demonstrates that a universal audio tokenizer, combined with the Zen language backbone (Qwen3-derived, 0.6B–32B dense plus a 30B-A3B MoE variant), supports a single model spanning ASR, multilingual recognition, and TTS synthesis. The streaming encoder's low first-token latency targets real-time conversational applications. The unified tokenizer eliminates the traditional pipeline approach, enabling cross-modal audio-visual-language reasoning within a single model.

References

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